The Naylor Model

v6 · Intuitive Outputs

Plain-English feature names · Statcast-style glossary
GLM table now reads in 'SB % Boost per Tier' instead of coef_z

Companion: Variable_Glossary.pdf

Executive Summary

Per-pitch data: 1,720,727 pitches w/ runner on 1st (2018-2026)
Attempts identified: 2,533
Qualified runner-seasons (SB+CS≥10): 1,467

What changed from v5

- ALL column names now plain English — no more coef_z, OR/SD, pct_in_HL.
- GLM table shows 'SB % Boost per Tier' = pp change in success rate
for a 1-SD improvement in the feature.
- Added pitcher pickoff-rate-faced and weak-arm-catcher share.
- Dropped pct_in_HL (was a Statcast data artifact — see v5 §9).

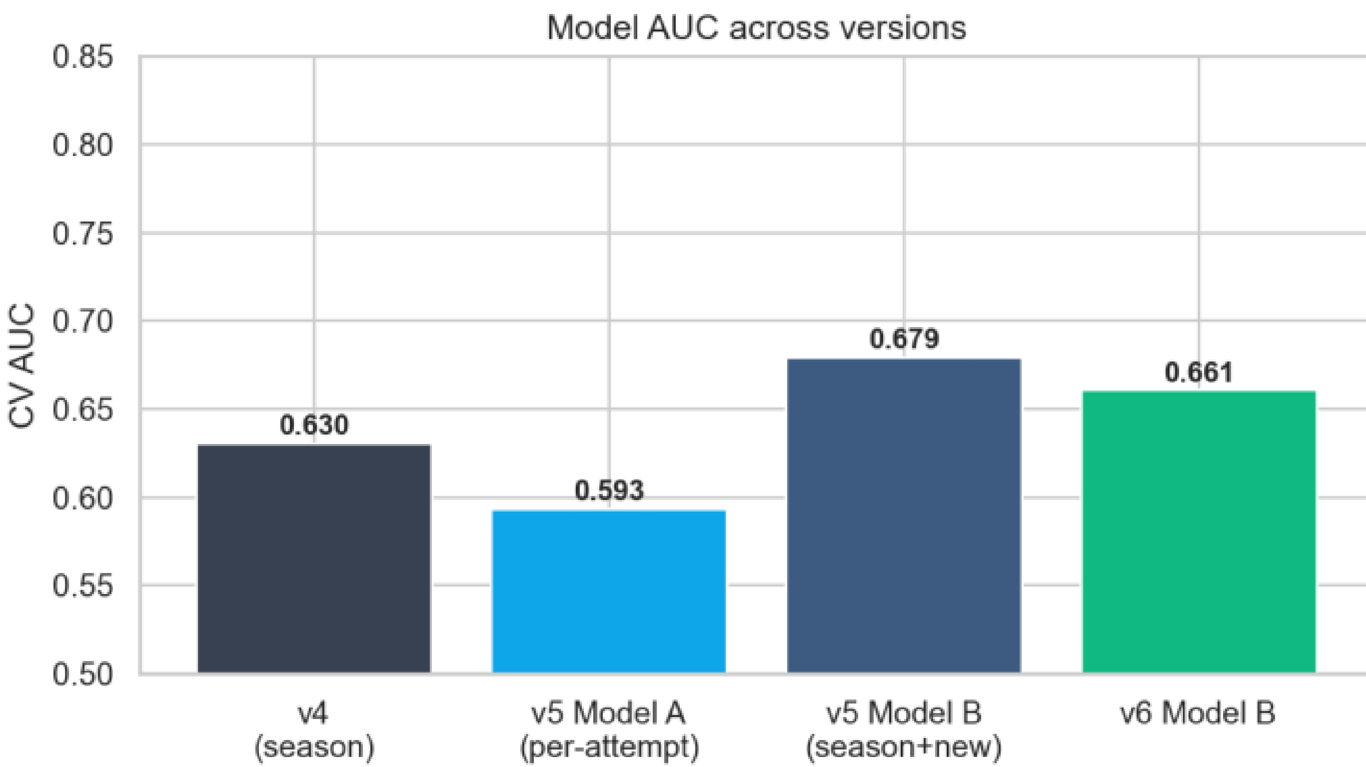
AUC summary

- v4 baseline: 0.6300
- v5 Model A (per-att): 0.5933
- v5 Model B (season): 0.6794
- v6 Model B (full): 0.6605
- v6 Model B (pre-23): 0.7194
- v6 Model B (post-23): 0.6924

Naylor + Soto rank under SSSI v6

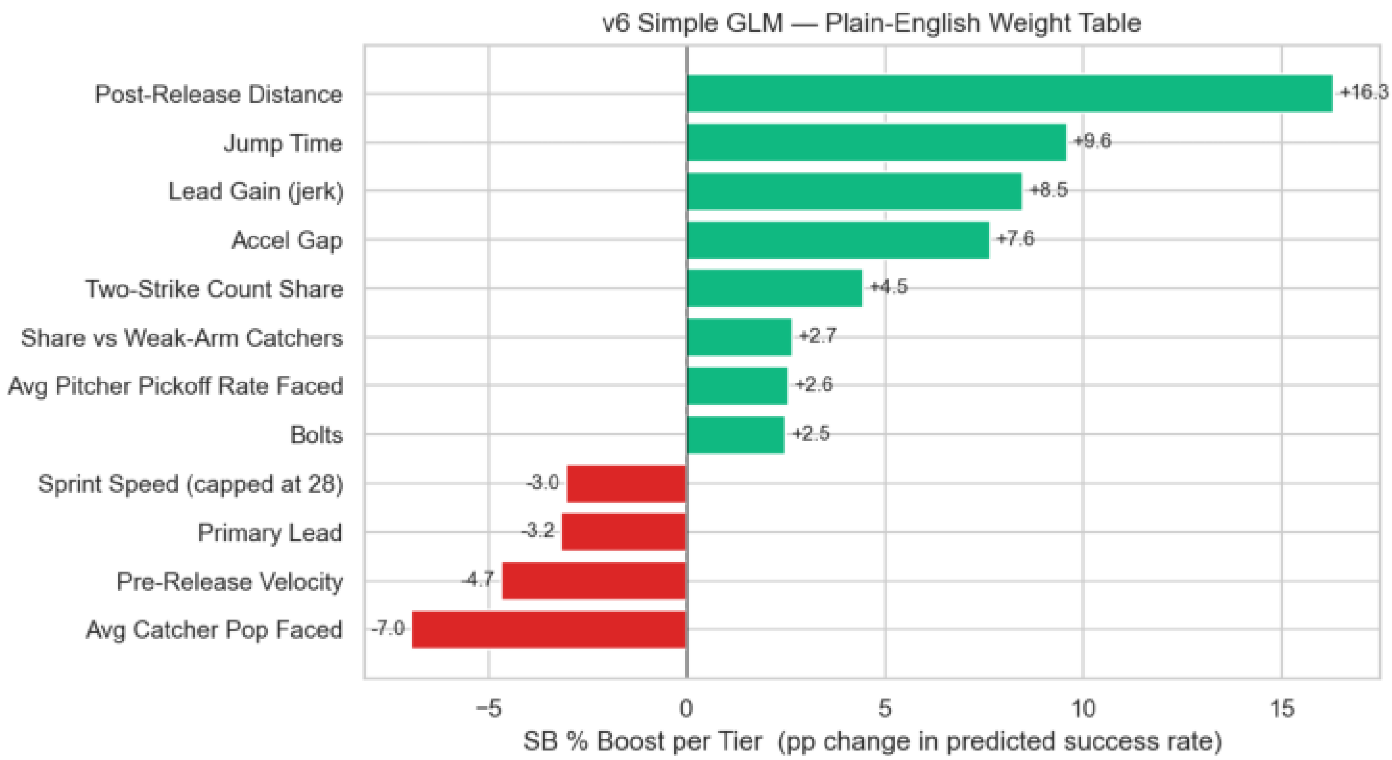
# 1	Josh Naylor	2025	SSSI +1.90	SB/CS 30/2	real_sb_pct 0.916
# 2	Josh Naylor	2026	SSSI +1.78	SB/CS 12/2	real_sb_pct 0.836
# 5	Juan Soto	2025	SSSI +1.21	SB/CS 38/4	real_sb_pct 0.891
# 14	Josh Naylor	2023	SSSI +1.02	SB/CS 10/3	real_sb_pct 0.772
# 76	Juan Soto	2023	SSSI +0.49	SB/CS 12/5	real_sb_pct 0.722
# 84	Juan Soto	2024	SSSI +0.45	SB/CS 7/4	real_sb_pct 0.681
#123	Juan Soto	2019	SSSI +0.29	SB/CS 12/1	real_sb_pct 0.883
#146	Juan Soto	2021	SSSI +0.23	SB/CS 9/7	real_sb_pct 0.614

§1 · AUC Comparison



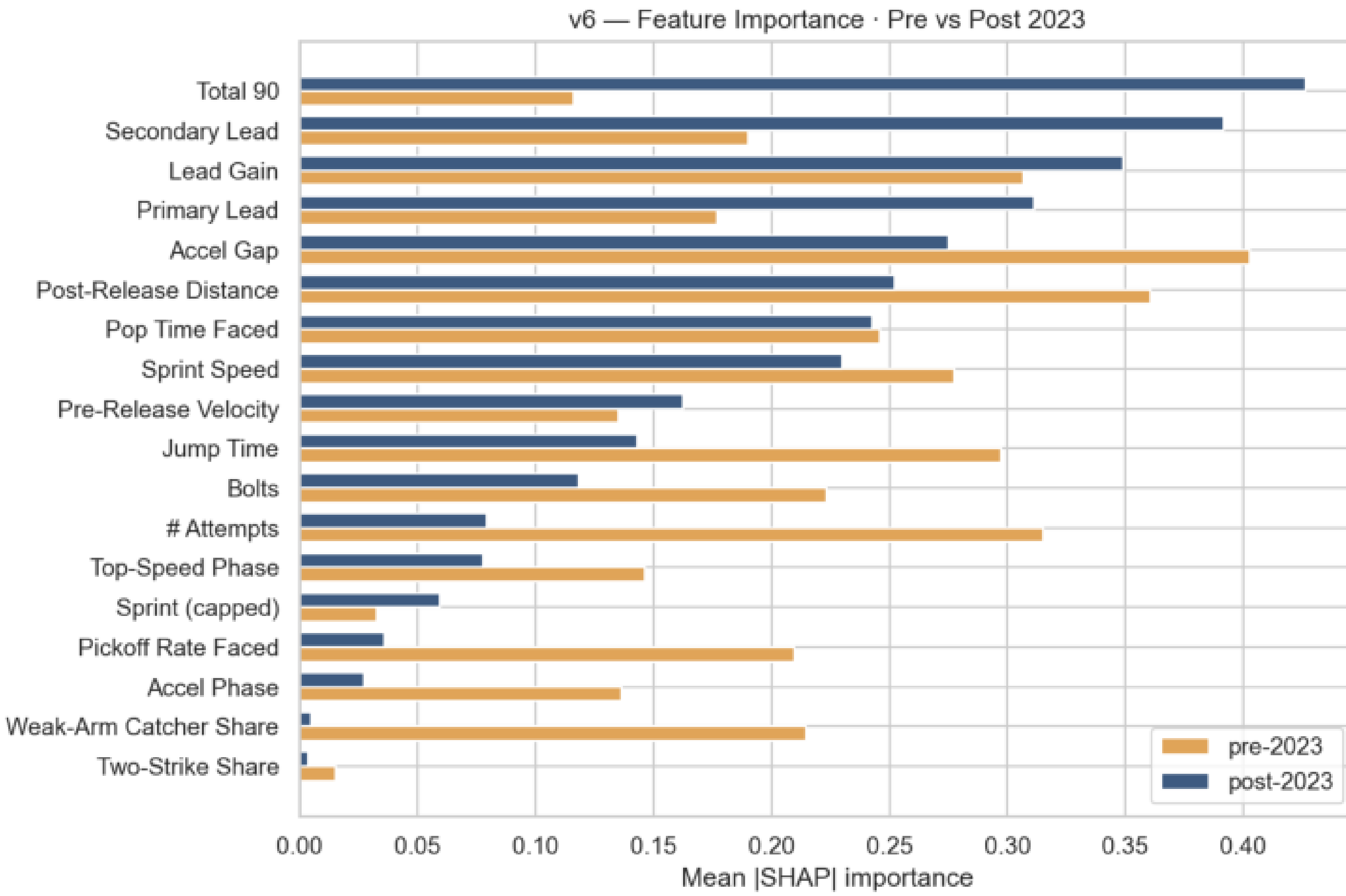
v6 Model B at the season level, with the v6 feature set including real catcher pop, pitcher pickoff rate, lead variables.

§2 · GLM Plain-English Weight Chart



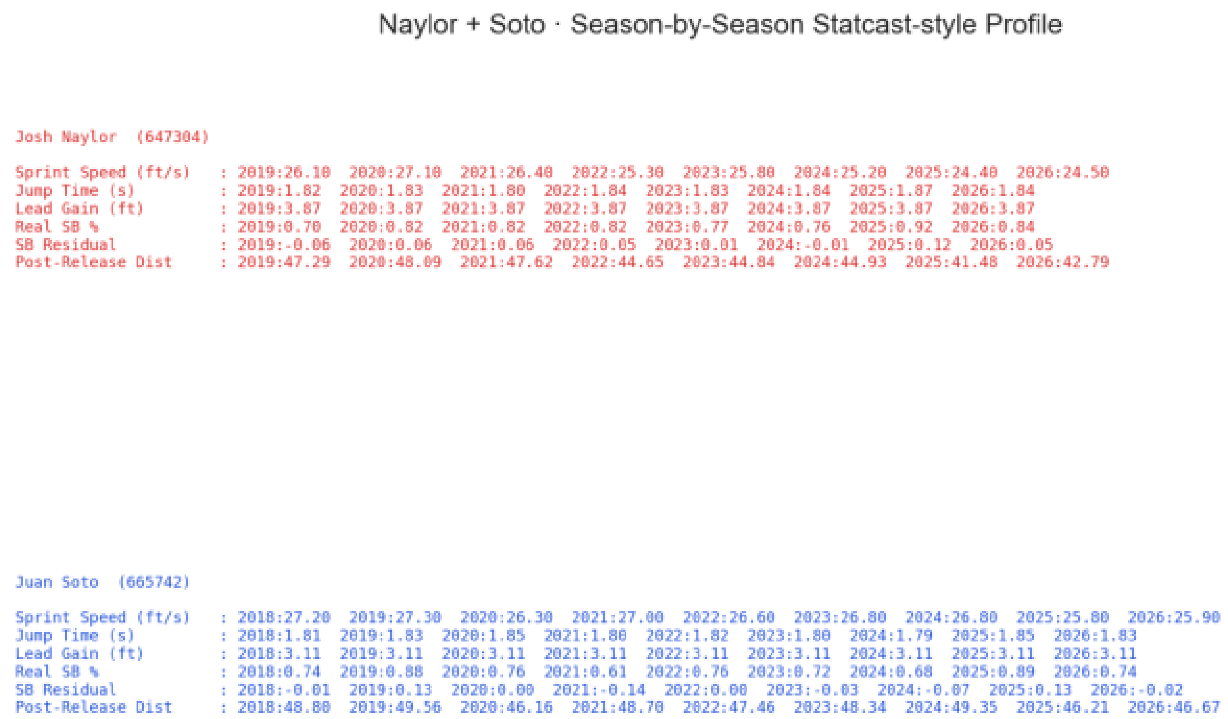
Each bar = predicted percentage-point change in SB success when the runner improves on that feature by ONE TIER (1 SD). Positive and green = the feature HELPS; red/orange = HURTS.

§3 · Feature Importance · Pre vs Post 2023



How feature importance shifted after the 2023 rule change.

§4 · Naylor + Soto Statcast-style Profile



Season-by-season values of the key features for the two archetypal slow-but-effective stealers.

§5 · GLM Weight Table — Plain English

Baseline P(success) at league-average inputs ≈ 0.607

Read each row as:
'If a runner improves [Feature] by 1 tier (1 SD), the model
expects their SB success rate to change by [Boost] percentage
points, equivalent to multiplying their odds by [Mult].'

Weight table (sorted by absolute boost)

Post-Release Distance	boost +16.32 pp	odds × 2.170	(higher_is_better = True)
Jump Time	boost +9.58 pp	odds × 1.531	(higher_is_better = False)
Lead Gain (jerk)	boost +8.47 pp	odds × 1.452	(higher_is_better = True)
Accel Gap	boost +7.64 pp	odds × 1.398	(higher_is_better = True)
Avg Catcher Pop Faced	boost -6.97 pp	odds × 0.752	(higher_is_better = True)
Pre-Release Velocity	boost -4.69 pp	odds × 0.824	(higher_is_better = True)
Two-Strike Count Share	boost +4.45 pp	odds × 1.211	(higher_is_better = False)
Primary Lead	boost -3.19 pp	odds × 0.876	(higher_is_better = True)
Sprint Speed (capped at 28)	boost -3.05 pp	odds × 0.881	(higher_is_better = True)
Share vs Weak-Arm Catchers	boost +2.66 pp	odds × 1.120	(higher_is_better = True)
Avg Pitcher Pickoff Rate Faced	boost +2.56 pp	odds × 1.115	(higher_is_better = False)
Bolts	boost +2.48 pp	odds × 1.111	(higher_is_better = True)

Technical notes (for stats readers)

boost_pp = sigmoid(intercept + coef) − sigmoid(intercept) × 100
odds_mul = exp(coef)
coef = standardised logistic-regression coefficient

Edit DF_v6_GLM_PlainEnglish.csv to hand-tune. No retraining needed.

§6 · SSSI v6 Top 15

Weights: sb_res=0.1 gap=0.05 gain=0.05 jump=0.0 prim=0.05 speed=-0.3 pre=0.0 post=0.0
(80% of runners used for grid search; Naylor + Soto held out.)

# 1	Josh Naylor	2025	SSSI +1.90	spd 24.4	jump 1.87	gain 3.87	SB/CS 30/2
# 2	Josh Naylor	2026	SSSI +1.78	spd 24.5	jump 1.84	gain 3.87	SB/CS 12/2
# 3	Freddie Freeman	2024	SSSI +1.26	spd 25.8	jump 1.81	gain 4.33	SB/CS 9/2
# 4	Christian Vázquez	2021	SSSI +1.22	spd 25.2	jump 1.93	gain 3.76	SB/CS 8/4
# 5	Juan Soto	2025	SSSI +1.21	spd 25.8	jump 1.85	gain 3.11	SB/CS 38/4
# 6	Kyle Schwarber	2025	SSSI +1.17	spd 25.9	jump 1.79	gain 4.43	SB/CS 10/2
# 7	Miguel Rojas	2024	SSSI +1.14	spd 25.8	jump 1.88	gain 4.37	SB/CS 8/2
# 8	Kyle Tucker	2024	SSSI +1.13	spd 26.0	jump 1.80	gain 3.01	SB/CS 11/0
# 9	Manny Machado	2024	SSSI +1.12	spd 25.8	jump 1.86	gain 3.31	SB/CS 11/2
# 10	Alec Burleson	2024	SSSI +1.11	spd 25.5	jump 1.88	gain 3.33	SB/CS 9/4
# 11	Ryan McMahon	2022	SSSI +1.09	spd 25.8	jump 1.79	gain 4.47	SB/CS 7/3
# 12	Manny Machado	2025	SSSI +1.08	spd 25.8	jump 1.88	gain 3.31	SB/CS 14/3
# 13	Joc Pederson	2024	SSSI +1.06	spd 25.5	jump 1.83	gain 3.28	SB/CS 7/4
# 14	Josh Naylor	2023	SSSI +1.02	spd 25.8	jump 1.83	gain 3.87	SB/CS 10/3
# 15	TJ Friedl	2024	SSSI +0.92	spd 26.5	jump 1.77	gain 3.88	SB/CS 9/1

§7 · Honest Data Limitations

What the model has access to:
- Sprint speed, running splits (real, per-runner-season, 2015+)
- Real catcher pop time (2018+, per-year)
- Real SB / CS counts (MLB Stats API)
- Career-aggregate lead profiles for runners and pitchers
- Per-pitch catcher_id, pitcher_id, balls, strikes, outs, inning

What the model does NOT have:
- Per-pitch pitcher delivery time (not publicly published)
- The EXACT count when an SB happened (we use AB's final count)
- Per-attempt lead distance (we use the runner's career mean)

Why the AUC ceiling exists:
- Stolen-base success has high intrinsic noise — pitch sequence, runner read, base-coach decisions, exact location at release.
- With public data the ceiling is around AUC 0.70 - 0.75.
- To push higher would require TrackMan / Hawk-Eye raw outputs.